

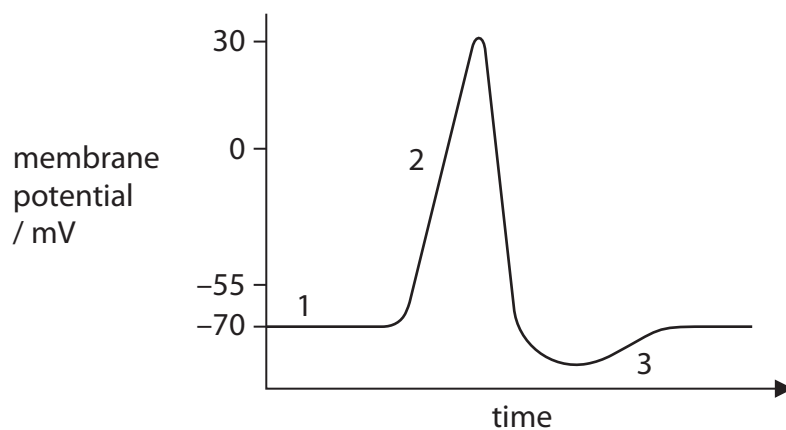
6 The nervous system is a complex collection of neurones and specialised cells that transmit impulses between different parts of the body.

(a) Which substance can transmit a nerve impulse from one neurone to the next one?

(1)

- ☐ **A** acetylcholine
- ☐ **B** ADP
- ☐ **C** cholesterol
- ☐ **D** NADP

(b) The diagram shows the change in membrane potential as an action potential is transmitted in a motor neurone.



Explain why the membrane potential changes between 1 and 2.

(2)

.....

.....

.....

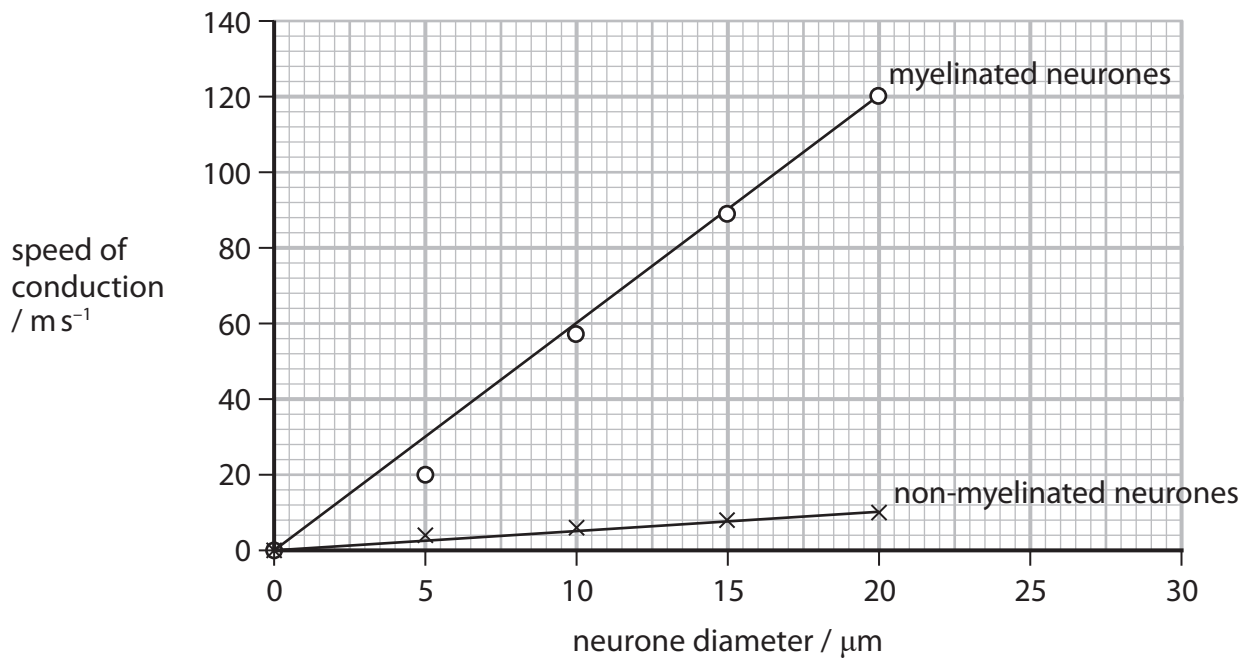
.....

.....

.....

(c) Myelinated neurones are surrounded by a fatty sheath.

The graph shows the speed of conduction in myelinated neurones and non-myelinated neurones.



- (i) Calculate the percentage difference in the gradient of the myelinated and non-myelinated neurones.

The gradient for non-myelinated neurones is 0.5.

The equation for a straight line is $y = mx + c$.

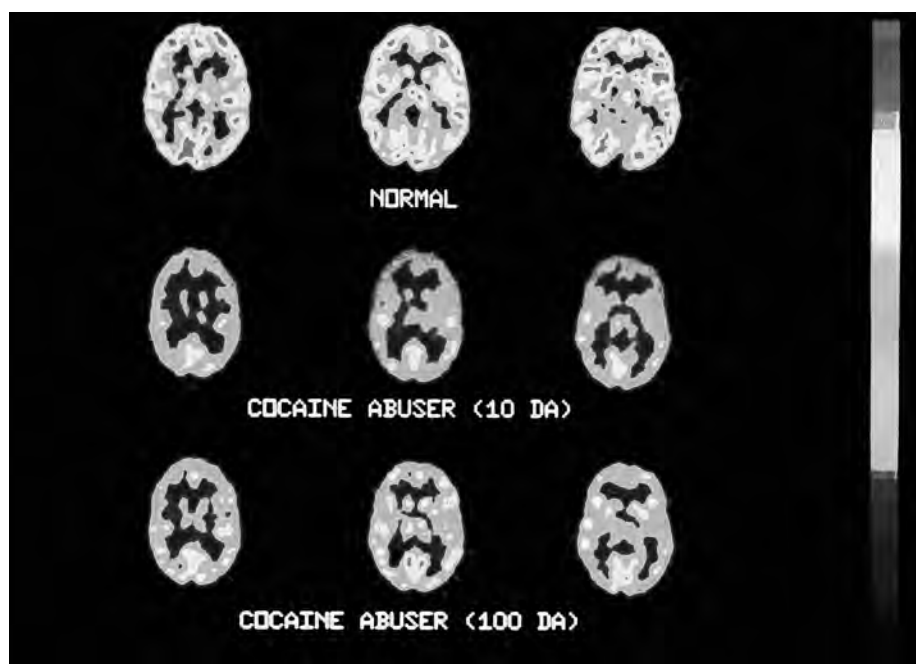
(3)

Answer%

(ii) Explain why the speed of conduction differs in myelinated and non-myelinated neurones.

(3)

- (d) The positron emission tomography (PET) images show the brain of an individual before (control) and after taking the drug cocaine.



(Source: © BROOKHAVEN NATIONAL LABORATORY / SCIENCE PHOTO LIBRARY)

Explain how positron emission tomography (PET) can be used to identify the change in activity in the brain after taking this drug.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question 6 = 13 marks)